

Term Paper Checkpoint 2: Question Development & CRAAP Analysis

Now that you are comfortable with the mechanics of performing a systematic literature search. It's time to start thinking about finding other data that doesn't exist in nicely formatted databases.

Before we can get to that, we need to figure out what your research questions are for your term paper. As with last checkpoint, you're not tied to these questions, you can always over-write your entire work each checkpoint, but the further we go, the more work it will be to throw everything out and start over. So this is a good opportunity to really think carefully about whether the topic you've chosen is something you're going to want to stick with for the rest of the term.

Research Questions:

It's important for a research paper to clearly define the questions that it is setting out to answer. By now, you (hopefully) have an idea of the topic you want to cover, and have done a bit of reading around the sub-topics and related topics. However, if you start out with just a vague notion of topic, you will wind up with an un-focused and sloppy paper.

You will need to refine your topic into 3-5 core research questions you are hoping to answer with your paper. The questions should be concrete enough that you have a chance of answering them, but open ended enough that there won't be a simple answer.

Examples of bad research questions:

- I'd like to research about dogs playing computer games (not a question)
- Has anyone ever tried to teach a dog to play computer games? (too simplistic)
- What happens when dogs play computer games? (too vague)
- What do dogs think about computer games? (not something that can be plausibly answered)

Examples of better research questions:

- What research has been done on the efficacy of using computer games to train service dogs?
- How do we measure human enjoyment of computer games? Is it possible to apply similar metrics to dogs?

- Is the current trend of “videogames for dogs” based on good quality research into canine behaviour?
- Are there physical or psychological harms that can be attributed to videogame playing in dogs?

Systematic Search (again)

Now that you have some solid research questions, you will likely need to at least modify your systematic search from TPC1. (If you lucked out into a search last time that still fits your questions. Congratulations, you can just copy-paste your old material into this section). So now you should re-run your systematic search from your previous checkpoint, with an updated query that will cover the research questions you’ve come up with. You will need to perform the same steps as you did in your previous checkpoint, and produce a similar report of query + databases + bibliography

Unsystematic Search

The odds are pretty good that you won’t be able to find everything you want to find about your topic in peer reviewed databases (If you can, then maybe you want to re-think your questions to expand into sub-topics like public perception or misinformation about your topic).

There’s no nice database of materials for this (though the library search does allow you to search newspapers and other periodicals which may be useful). So you will need to use your intuition to find material that is of high enough quality to be worth your time, but covers things like opinions, perceptions, official positions of organizations/individuals, etc.

You will want to find a set of 10-20 documents that seem (at first glance) to be relevant to your topic/research questions. As with your systematic search, you will need to present a bibliography of those items (hint: If you are unsure how to cite things that aren’t research papers, the library has good guides and even links to citation generators (<https://guides.library.utoronto.ca/citing>))

CRAAP Analysis

Now that you’ve got a good set of resources to start from (10-20 from your systematic search + 10-20 from your unsystematic search), it’s time to actually read them.

At this point, you don’t need to go into the level of depth you’ll need for your final paper. A quick skim of the paper + focus on a few key areas such as methodologies or author background should be enough. But if you’ve got the time now (reading week), this could be a good opportunity to read the papers in more detail and take notes (you won’t remember everything by the time you’re writing your term paper).

For each resource, you'll need to do a quick CRAAP analysis. Some papers may not have relevant information on each component of CRAAP, but some may require more detail.

An example might look something like this:

Madeupington, Fakename. "The computer game that will teach your dog to fetch" Dog Training Monthly, www.dogtrainingmonthly.com/news/computergame.html. Accessed 1 Jan 2024.

- Currency - Article was this year, but based on a study from 2020. FetchQuest 2 has since been updated to FetchQuest 3, but most discussed features seem to still be relevant
- Relevance - Directly speaking to research question 3
- Authority - Author is regular contributor to dogtrainingmonthly, referencing research done by internal FetchQuest team with no author attribution
- Accuracy - The conclusions are based on data that has not been made public. Only summary results are provided, so it is not possible to validate the statistical analysis provided
- Purpose - Author does not have obvious bias towards game company, but is referencing research presented by game company uncritically. No disclosure of paid journalism, but FetchQuest is an advertiser on dogtrainingmonthly

What to Submit

In a single document, you must submit:

- Your research questions
- Your systematic search (even if un-updated from your previous version in TPC1)
- Your unsystematic search
- Your CRAAP analysis for at least 10 documents (at least 5 from systematic and 5 from unsystematic search)